

Block_Cipher	Structure	Standard	Block_Length	CPU_Clock_Freq	Throughput	Total_Clock_Cycles	Util_Memory	Security_Level	Known_Vulnerabilities
3DES	Feistel	No	64	-	1.76	3600	20.7	Medium	Vulnerable to meet-in-the-middle attacks and block collision attacks; exploited by the Sweet32 attack (CVE-2016-2183).
3-Way	SPN	No	96	66	2	-	-	Medium	Vulnerable to related-key cryptanalysis; can be broken with one related key query and about 2 ²² chosen plaintexts.
μ ²	Feistel	No	64	3500	148.28	-	-	-	-
AES-128	SPN	Yes	128	206	72.84	362	20	High	Related-key attacks and known-key attacks (only against improperly configured systems or reduced-round versions).
Akellarre	Lai-Massey	No	128	130	3.22	-	-	Low	Susceptible to ciphertext-only attacks; weak round function preserves input parity, insecure against chosen plaintext attacks.
Anubis	SPN	No	128	1333	1160.71	147	-	High	-
ARIA	SPN	No	128	14.74	0.28	6624	-	High	Susceptible to impossible differential cryptanalysis and meet-in-the-middle attacks on reduced rounds.
BaseKing	SPN	No	128	-	-	1949	1.7	Medium	Vulnerable to related-key attack and power analysis.
Blowfish	Feistel	No	64	-	4.28	-	9.38	High	Vulnerable to Sweet32 birthday attacks; reduced-round variants susceptible to known-plaintext attacks on weak keys.
Camellia	Feistel	Yes	128	667	251.85	339	3.07	High	Vulnerable to cache timing attacks.
CAST-128	Feistel	No	64	16	0.036	16000	-	Medium	Susceptible to differential cryptanalysis, linear cryptanalysis, and weaknesses related to the nonsurjectivity of the f-function.
CAST-256	Feistel	No	128	200	37.8	-	-	High	Minor weaknesses such as non-surjective attacks and HOD attacks, but nothing extendable beyond 5-6 rounds.
CLEFIA	Feistel	No	128	2400	1480	206	8	High	Susceptible to impossible differential attacks, particularly on reduced rounds.
CS-Cipher	Feistel	No	64	133	8.34	973	-	-	-
CRYPTON	SPN	No	128	200	6.4	390	-	Medium	Susceptible to partitioning cryptanalysis and weaknesses in key schedule.
DES	Feistel	No	64	100	0.052	124000	0.5	Low	Susceptible to brute-force attacks, differential cryptanalysis, and linear cryptanalysis.
Hierocrypt	Nested SPN	No	128	599	48	-	-	High	Weaknesses in the key schedule; integral attacks against reduced-round variants (3.5 rounds).
ICE	Feistel	No	64	100	52.03	123	-	High	Differential cryptanalysis can break 15 out of 16 rounds with complexity 2 ⁵⁶ .
IDEA	Lai-Massey	No	64	667	22.61	1888	-	High	Susceptible to implementation issues such as side-channel attacks; weak keys in earlier versions.
KHAZAD	SPN	No	64	550	86	409	-	Low	Susceptible to attacks on 5 out of 8 rounds; best known attack breaks 5 rounds faster than exhaustive key search.
KLEIN-64/80/96	SPN	No	64	1	0.03	1970	3.2	Medium	Susceptible to chosen-plaintext key-recovery attacks (up to 8 rounds of KLEIN-64) and potential related-key weaknesses.
LBlock	Feistel	No	64	-	-	3955	-	Medium	Susceptible to differential cryptanalysis, linear cryptanalysis, impossible differential cryptanalysis, and related-key attacks.
LED-64/128	SPN	No	64,128	1600	224.56/148.84	456/688	-	High	Susceptible to differential and linear cryptanalysis.
LOKI97	Feistel	No	128	233	1	-	-	Medium	Susceptible to linear cryptanalysis requiring 2 ⁵⁶ known plaintexts.
MAGENTA	Feistel	No	128	200	1.08	23694	-	Low	Susceptible to various cryptanalysis methods; weaknesses discovered during AES evaluation.
MARS	Type-3 Feistel	No	128	200	85	300	2.16	High	-
Mercy	Feistel	No	4096	233	207.11	4608	1.5	Low	Susceptible to differential attacks, specifically Scott Fluhrer's differential attack breaks the cipher.
MESH-64/96/128	Lai-Massey	No	64,96,128	667	20.7/16.5/15.4	2054/3869/5536	-	High	-
NOEKEON	ARX	No	128	28.56	7.7	475	3.68	High	Vulnerable to certain peculiar kinds of related-key cryptanalysis.
RECTANGLE	SPN	No	64	2500	655.74	244	3	Medium	Can be attacked with a shortcut attack on 18 rounds out of 25 rounds.
RC5	Feistel-like	No	64	50	1.2	-	-	High	Susceptible to differential and linear cryptanalysis; vulnerable to brute-force attacks if key size is insufficient.
RC6	Feistel	No	128	254	12.6	148	-	High	Susceptible to certain cryptanalytic attacks.
SC2000 (128-bit)	SPN and Feistel	No	128	500	268.91	238	1.6	High	Potentially vulnerable to physical attacks
Serpent	SPN	No	128	133	0.0089	-	-	High	Susceptible to attacks on reduced rounds (11 or 12 rounds) with significant known plaintexts; full 32-round version remains secure against known attacks.
SHARK	SPN	No	64	266	6.3	-	-	Medium	Susceptible to interpolation attacks.
SIMON	SPN and Feistel	Yes	64	14.74	17.71	10656	0.19	High	Susceptible to differential cryptanalysis; not designed to resist known-key distinguishing attacks.
SPECK	SPN and Feistel	Yes	64	14.74	28.08	6720	0.16	Medium	Susceptible to differential cryptanalysis and machine learning-based attacks.
Square	SPN	No	128	100	2.63	-	-	Medium	Vulnerable to Square attack
TWINE	Feistel	No	64	1	0.03	2168	1.72	Medium	Susceptible to slide attacks and biclique cryptanalysis due to low diffusion properties.